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RESEARCH

Molecular and genetic analysis of methicillin-resistant *Staphylococcus aureus* (MRSA) in a tertiary care hospital in Saudi Arabia

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Abstract
Methicillin-resistant *Staphylococcus aureus* (MRSA) continues to pose significant challenges in healthcare settings due to its multi-drug resistance (MDR) and virulence. This retrospective study examines the molecular and resistance profiles of MRSA isolates from a tertiary care hospital in Saudi Arabia, providing valuable insights into regional epidemiology. A total of 190 MRSA strains were analysed to assess antimicrobial susceptibility, genetic diversity, and virulence factors. Antimicrobial susceptibility testing was conducted according to CLSI guidelines, while molecular characterization involved *spa* typing, SCCmec typing, and DNA microarray analysis to determine clonal complexes (CCs), resistance genes, and virulence determinants. The isolates showed extensive resistance to beta-lactam antibiotics, with 78% classified as MDR. Notably, resistance to fusidic acid and ciprofloxacin was detected in 70% and 55% of isolates, respectively. The most prevalent clonal complexes—CC5, CC6, and CC22—comprised over 60% of the isolates and exhibited diverse *spa* types. The Panton-Valentine leukocidin (*PVL*) gene, linked to heightened virulence, was identified in approximately 20% of isolates, particularly within CC5, CC30, and CC80. Enterotoxin genes (*sea* and *seb*) and immune evasion genes (*sak*, *chp*, and *scn*) were also commonly detected, reflecting the isolates' capacity to adapt and persist within the hospital environment. These findings underscore the high burden of MDR MRSA with considerable genetic diversity and virulence potential. The study highlights the urgent need for strengthened molecular surveillance and targeted infection control measures to limit MRSA transmission and effectively manage infection risks in healthcare facilities.

Keywords Clonal complexes · Methicillin-resistant *Staphylococcus aureus* · Multi-drug resistance · Virulence factors · Tertiary care hospital

Abbreviations

ATCC	American Type Culture Collection
CA-MRSA	Community-associated MRSA
CC	Clonal complex
CLSI	Clinical and Laboratory Standards Institute
HA-MRSA	Healthcare-associated MRSA
HLB	Human leukocyte bacteria (related to immune-evasion genes like *sak*, *chp*, *scn*)
MRSA	Methicillin-resistant *Staphylococcus aureus*
MIC	Minimum inhibitory concentration

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MDR	Multi-drug resistance
PCR	Polymerase chain reaction
PVL	Panton-Valentine leucocidin
SCCmec	Staphylococcal cassette chromosome mec

Introduction

Methicillin-resistant *Staphylococcus aureus* (MRSA) is a prominent cause of healthcare-associated infections worldwide and remains a critical public health challenge due to its high morbidity and mortality rates, multi-drug resistance (MDR), and ability to adapt across diverse environments (Green et al. 2012; Lee et al. 2018; Turner et al. 2019; Wang et al. 2022; Abebe and Birhanu 2023). First identified in the 1960s, MRSA has evolved into a complex pathogen with distinct healthcare-associated (HA) and community-associated (CA) lineages, each demonstrating unique genetic characteristics and resistance profiles (Uhlemann et al. 2014; Njenga et al. 2022; Shoaib et al. 2022). In recent years, CA-MRSA strains have shown an alarming propensity for spreading within healthcare settings, presenting significant infection control challenges (Raygada and Levine 2009; Samuel et al. 2023).

In the Gulf Cooperation Council (GCC) region, methicillin resistance rates among *Staphylococcus aureus* isolates show notable variability. The United Arab Emirates (UAE) and Qatar reported lower rates than Oman and the Kingdom of Saudi Arabia (KSA). A study from Dubai recorded a 29% resistance rate among 62 *S. aureus* isolates linked to purulent skin and soft tissue infections (SSTIs) (Al Jalaf et al. 2018).

In the Levant and North Africa, resistance rates were more variable. Lebanon and Tunisia generally reported lower resistance rates. However, a Lebanese study found a high resistance rate of 72% (Tokajian et al. 2010). In contrast, a larger survey of 4890 isolates across 16 hospitals in Lebanon reported a lower resistance rate of 28%, highlighting the impact of sample size and methodology (Chamoun et al. 2016).

Iraq showed extremely high resistance rates, particularly in combat support hospitals treating US military personnel and in burn care centres, where rates reached 88% in SSTIs (Co et al. 2011; Roberts and Kazragis 2009). In North Africa, Algeria reported alarmingly high rates, including 86% in diabetic foot infections in Annaba (Djahmi et al. 2013) and 75% in postoperative surgical wound infections in Tlemcen (Rebiahi et al. 2011). By comparison, a study from Algiers reported a much lower rate of 19%, demonstrating regional variation within the country (Djoudi et al. 2013). Tunisia reported lower rates, including 18% among neutropenic patients, notably lower than most studies from Egypt and Algeria (Bouchami et al. 2009).

Few studies differentiated *S. aureus* resistance rates based on the origin of infection, specifically community-onset MRSA (CO-MRSA) versus hospital-onset MRSA (HO-MRSA). In the KSA, CO-MRSA resistance was reported at 51% (Abdel-Fattah 2005). In Egypt, CO-MRSA rates ranged from 12 to 47% (Sobhy et al. 2012; Abdel-Maksoud et al. 2016), while in Algeria, rates varied between 24 and 40% (Alioua et al. 2014; Rebiahi et al. 2011). Hospital-onset MRSA showed consistently higher resistance rates. In the KSA, resistance reached 66% (Abdel-Fattah 2005). Egypt reported rates as high as 77% (Sobhy et al. 2012; Abdel-Maksoud et al. 2016), while Algeria's rates ranged from 47 to 75% % (Alioua et al. 2014; Rebiahi et al. 2011). These findings underscore the higher resistance burden in hospital settings and the significant role of acquisition sites in MRSA epidemiology.

The global healthcare burden posed by MRSA is evident from its resistance to numerous antibacterial drug classes, including beta-lactams, macrolides, and fluoroquinolones (Raygada and Levine 2009; Vestergaard et al. 2019; Ali Alghamdi et al. 2023). With few remaining treatment options, the reliance on last-resort antibacterial drugs such as vancomycin and teicoplanin has increased, although reduced susceptibility to these agents has also been documented in several studies (Reinseth et al. 2020). Multi-drug resistance, particularly among HA-MRSA isolates, compounds the challenges of managing MRSA infections, as these strains frequently harbour a range of antibacterial drug-resistance genes (Guo et al. 2020; AlSaleh et al. 2023). The emergence of fusidic acid resistance in MRSA observed in 70% of isolates in this study, exemplifies this trend, and raises concerns regarding the effectiveness of common therapeutic regimens in local healthcare settings (Zhao et al. 2021). Saudi Arabia, as a high-resource Middle Eastern country, faces unique healthcare dynamics with its mix of domestic and international patient populations, which contribute to pathogen diversity in its hospitals (Mate et al. 2017). Despite numerous studies on MRSA epidemiology, there remains a paucity of research focusing on the genetic characteristics of MRSA in the GCC region, where population mobility and healthcare standards may influence pathogen spread. This study addresses this gap by examining the clonal composition, resistance profiles, and virulence determinants of MRSA strains isolated in a tertiary hospital setting. By employing molecular techniques such as s*pa* typing, SCCmec typing, and microarray analysis, we aim to elucidate the genetic architecture and resistance mechanisms underpinning MRSA epidemiology in the region (Moosavian et al. 2020).

The genetic diversity of MRSA, particularly concerning *spa* types and clonal complexes (CCs), is an area of increasing interest due to its implications for transmission dynamics and strain adaptability (Shoaib et al. 2022). Global studies highlight certain clonal lineages, including CC5, CC6, and CC22, as common to both healthcare and community environments (Bispo et al. 2020). Furthermore, this study highlights the necessity of continuous genomic surveillance to monitor for emerging resistance patterns and inform effective public health responses.

In summary, this study aims to advance knowledge of MRSA epidemiology within a Middle Eastern healthcare context, identifying isolates' genetic and resistance profiles from a tertiary hospital. This research adds valuable data to the global discourse on MRSA, facilitating more targeted infection control strategies and improving clinical outcomes by exploring the prevalence of specific clonal complexes, *spa* types, and resistance genes.

Material and methods

Study design and MRSA isolates

A retrospective study included 190 MRSA strains obtained from King Khalid Hospital (KKH), Hail, Saudi Arabia. Initial identification of MRSA was performed through Gram staining, coagulase testing, and the BD Phoenix M50, which classified isolates based on biochemical properties (Cartwright et al. 2013). Isolates were subsequently cultured on brain-heart infusion agar (BHIA) and incubated at 35°C for 18–24 h. For long-term preservation, we use glycerol stock. Here, we suspended pure colonies in 15% sterile glycerol prepared in trypticase soy broth (TSB) with 15% glycerol and stored at −20°C.

Ethical considerations

The study was conducted following the Declaration of Helsinki and approved by the Institutional Ethics Committee of the Hail Health Cluster (Approval No: H-08-L074) and the University of Hail (Approval No: H-2023-390), Saudi Arabia. Confidentiality and anonymity were strictly maintained throughout.

Antibacterial susceptibility profile of the MRSA isolates

Antibacterial drug susceptibility testing was performed using the Kirby–Bauer disc diffusion method as per CLSI (2020) guidelines, complemented by the BD Phoenix M50 system. This approach ensures high reliability, reproducibility, and adherence to international standards, providing comprehensive and accurate results for antibacterial susceptibility testing.

Isolates were assessed against a range of antibacterial drugs, including penicillin (10 IU), cefoxitin (30 µg), gentamicin (10 µg), erythromycin (15 µg), ciprofloxacin (5 µg), and tetracycline (10 µg). The minimum inhibitory concentrations (MICs) for vancomycin and teicoplanin were determined using E-test strips (bioMérieux) following the manufacturer's instructions (Sancak et al. 2013). MIC breakpoints were interpreted according to CLSI standards, and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines 2020 were consulted where necessary, particularly for fusidic acid (Gaur et al. 2023). Quality control for testing was maintained using *Staphylococcus aureus* ATCC 25923 and ATCC 29213 as control strains (Treangen et al. 2014).

DNA extraction for PCR testing

To prepare DNA for molecular analysis, bacterial cells were cultured overnight on BHIA at 35°C. A few colonies (3–5) from the fresh culture were transferred into a microcentrifuge tube with 50 µL of lysostaphin and RNase solution and incubated at 37°C for 20 min (Farhangnia et al. 2014). The cells were then treated with proteinase K (20 mg/mL) and Tris buffer (0.1 M), followed by incubation at 60°C to facilitate cell lysis (Hsueh et al. 2023). This was followed by heat inactivation at 95°C for 10 min to deactivate proteinase K. DNA extraction was performed using QIAmp DNA Mini Kit (QIAGEN, Hilden, Germany). The quality and concentration of the extracted DNA were assessed using a NanoDrop Spectrophotometer (Thermo Fisher Scientific, USA) by evaluating the A260/280 and A260/230 ratios to confirm purity. The DNA integrity was further verified through agarose gel electrophoresis, which demonstrated intact DNA with no signs of degradation. The prepared DNA was stored at 4°C for subsequent PCR amplification.

Genotyping of the MRSA isolates

For *spa* typing, the *spa* gene was amplified using specific primers. The PCR cycling protocol included an initial denaturation at 94°C for 4 min, followed by 25 cycles of denaturation at 94°C for 1 min, annealing at 56°C for 1 min, and extension at 72°C for 3 min (Lorenz 2012). A final extension step at 72°C for 5 min completed the reaction. PCR products were purified using ExoSAP-IT, and sequencing was performed using an automated genetic analyser. The resulting sequences were analysed with Ridom StaphType software to assign *spa* types to each isolate (Bell 2008).

Detection of virulence genes

The virulence genes among the MRSA isolates were detected using polymerase chain reaction (PCR). Specific primers targeting key virulence-associated genes were designed or selected based on previously published protocols (Hoseini Alfatemi et al. 2014). The genes included in the analysis were *lukS-PV/lukF-PV* (Panton-Valentine leukocidin), *sea*, *seb* (staphylococcal enterotoxins A and B), *tst1* (toxic shock syndrome toxin-1), and immune evasion genes *sak* (staphylokinase), *chp* (chemotaxis inhibitory protein), and *scn* (staphylococcal complement inhibitor) (Amissah et al. 2017).

DNA microarray analysis

Genetic profiling of MRSA isolates was conducted with a DNA microarray kit (INTER-ARRAY, Gmbh, Germany, Genotyping Kit for *S. aureus*), following the manufacturer's protocol (Alfouzan et al. 2023). This method enabled the identification of CCs and the detection of antibacterial drug resistance and virulence genes. Analysis of the hybridization results provided insights into the genetic background of each MRSA isolates, supporting classification into clonal complexes and assessment of resistance mechanisms and virulence factors (Abd El-Hamid et al. 2022).

Results

Distribution of MRSA isolates among patients

Among the 190 MRSA isolates obtained, 58% (n = 110) were from male patients, and 42% (n = 80) were from female patients. Specimens were collected across various departments of the tertiary care hospital, including intensive care units, general medical and surgical wards, and outpatient clinics. The majority of isolates were derived from blood samples (36%, n = 68), followed by skin and soft tissue samples (30%, n = 57), urine (12%, n = 23), respiratory specimens (10%, n = 19), and other sites, including wound and nasal swabs (12%, n = 23).

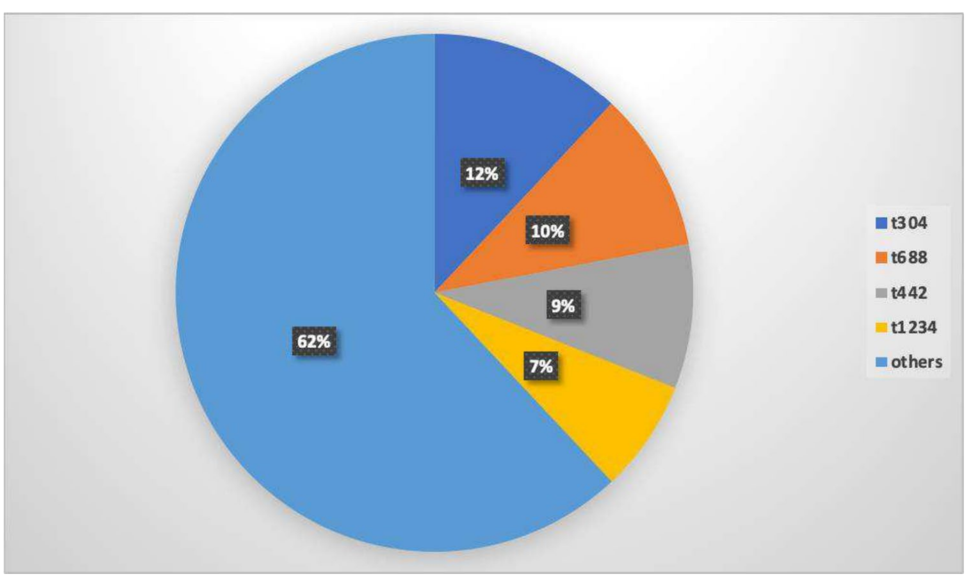
Antimicrobial susceptibility profile of the MRSA isolates

All MRSA isolates exhibited resistance to beta-lactam antibacterial drugs, confirmed by cefoxitin resistance. Susceptibility testing revealed that 90% of isolates were susceptible to vancomycin (MIC ≤ 2 μg/mL) and 92% to teicoplanin, with the remaining isolates displaying reduced susceptibility (MIC 2–4 μg/mL). Resistance rates to other antibacterial drugs were as follows: fusidic acid (70%), ciprofloxacin (55%), erythromycin (48%), gentamicin (25%), kanamycin (20%), and clindamycin (16%). Multi-drug resistance, defined as resistance to three or more classes of antibacterial drugs, was observed in 78% of the isolates, with combinations most frequently including fusidic acid, erythromycin, and ciprofloxacin.

Genetic analysis and *Spa* typing

Figure 1 highlights the diversity and frequency of predominant *spa* types among the MRSA isolates in this study. The most common *spa* type, t304, accounts for 12% of isolates (around 23 out of 190), followed closely by t688 at 10% (approximately 19 isolates), both of which are frequently observed in community-acquired MRSA (CA-MRSA) strains, suggesting their significant role in transmission. Type t442, found in 9% of the isolates (about 17 samples), shows lineage diversity, as it is commonly seen in both healthcare-associated and community settings. Meanwhile, t1234, representing 7% of the isolates (around 13 samples), appears less commonly but retains a presence in clinical

Fig. 1 Percentage distribution of predominant *spa* types among MRSA isolates



MRSA populations, often indicating sporadic incidence. The “Others” category, which combines various lower-prevalence *spa* types, makes up 62% of the isolates and reflects substantial genetic variability, possibly indicating different community or hospital introductions. This distribution provides insight into both dominant and sporadic MRSA lineages within the patient population.

Distribution of SCCmec types and CCs

Figure 2 illustrates the distribution of CCs among MRSA isolates from 190 patients. The most prevalent CC is CC5, representing 30% of the isolates, followed by CC6 at 18% and CC22 at 14%, indicating these as the dominant clonal lineages in this study population. CC97 and CC15 contribute 12% and 8%, respectively, while CC30 and CC80 make up 7% and 6%, highlighting a notable variety of MRSA clonal backgrounds. The “Others” category, encompassing less common CCs, accounts for the remaining 5% of isolates. This diversity underscores both the prevalence of established MRSA clones and the presence of sporadic lineages within the hospital setting.

The study identified multiple clonal complexes (CCs) among the MRSA isolates, with CC5, CC6, and CC22 being the most prevalent. Each of these complexes exhibited unique resistance profiles and genetic characteristics (Table 1). Specifically, *spa* typing revealed dominant types such as t304 in CC5, t688 in CC6, and t442 in CC22. A range of less common clonal complexes, including CC15, CC30, and CC80, were also detected, along with various *spa* types, demonstrating the genetic diversity within the MRSA population.

Spa types and genetic lineage diversity

The high prevalence of CC5, CC6, and CC22 aligns with trends observed in other studies, where CA-MRSA-associated lineages such as CC5 and CC6 have increasingly supplanted traditional healthcare-associated (HA) strains. Additionally, the wide distribution of *spa* types, including dominant and less common variants, highlights the genetic variability within MRSA isolates. These findings suggest that the genetic adaptability of MRSA enables it to persist and thrive in both community and healthcare settings. The detection of multiple less common *spa* types within the “Others” category may point to the possible introduction of new or sporadic MRSA lineages within the hospital environment.

Resistance profiles and key genotypes

The resistance profiles of these MRSA isolates reveal widespread multi-drug resistance, particularly across complexes CC5, CC22, and CC97. Resistance to fusidic acid is significant, with 70% of the isolates harbouring the *fusC* gene, a common fusidic acid resistance marker associated with SCCmec elements and often found in both CA-MRSA and HA-MRSA strains. Additionally, the *ermC* gene, conferring macrolide resistance, was highly prevalent among CC5 and CC22 isolates. Notably, CC22, which has been linked to the EMRSA-15 lineage, was also resistant to trimethoprim

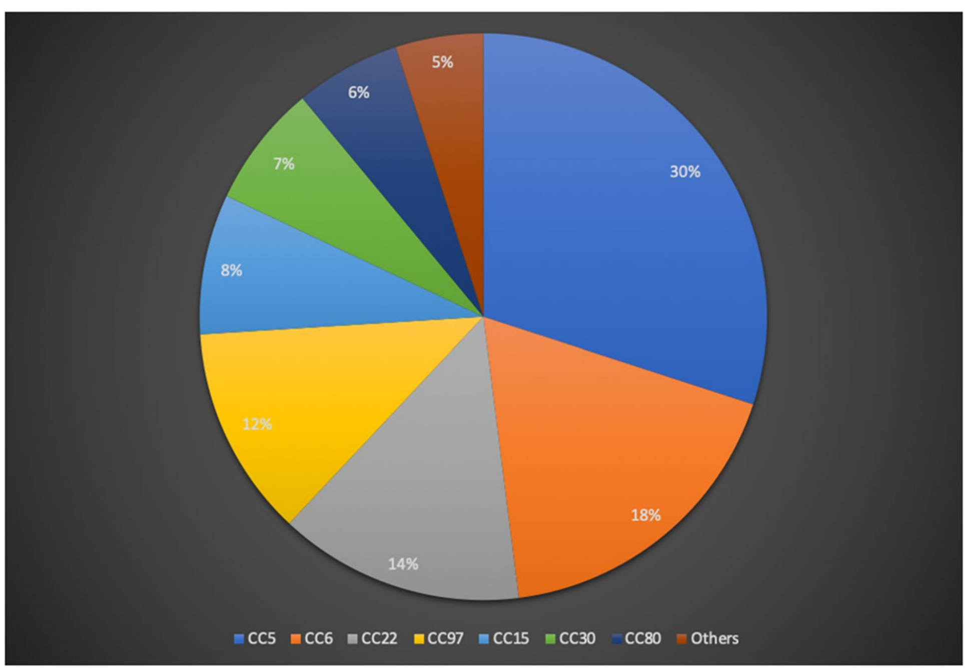


Fig. 2 Percentage distribution of clonal complexes (CC) among MRSA isolates

Table 1 Molecular characteristics of MRSA isolates (clonal complex, strain type, resistance genotype, and virulence factors)

Clonal complex (CC)	Strain type	*Spa* type	Antibacterial drug resistance genotype	Toxins	HLB-converting phages	Misc. genes
CC5	MRSA-IV [*PVL*+]	t304	*ermC, fusC, tetM*	*PVL, egc*	*sak, scn*	*agrII, cap5*
	MRSA-V + *SCCfus*	t1090	*blaZ, tetM, fusC*	*seb, sea*	*sak, scn*	*agrII, cap5*
	MRSA-VI	t311	*ermC, dfrS1*	*egc, etA*	*scn*	*agrII*
CC6	MRSA-IV	t688	*ermC*	*Sea*	*sak, chp*	*agrI, cap8*
	MRSA-IV + *SCCfus*	t701	*fusC*	*seb, sek*	*sak*	*agrI*
CC22	MRSA-IV [*tst1*+]	t442	*dfrS1, ermC, fusC*	*tst1, egc, sea*	*chp, scn*	*agrI, cap5*
	MRSA-IV	t5168	*dfrS1, tetK*	*tst1, seb*	*sak, scn*	*agrI*
	MRSA-IV	t2336	*fusC, tetK*	*tst1*	*sak, scn*	*agrI*
CC97	MRSA-V [*fusC*+]	t1234	*fusC, aacA-aphD*	*seb, sea , egc*	*sak, scn*	*agrI, cap5*
	MRSA-VI	t688	*dfrS1, tetK*	*seb, egc*	*sak, scn*	*agrI*
CC15	MRSA-V + *SCCfus*	t701	*fusC, tetK, blaZ*	*egc, sed*	*scn*	*agrII*
	MRSA-IV	t450	*fusC, ermC*	*egc, seb*	*sak, scn*	*agrII*
CC30	MRSA-IV [*PVL*+/*tst1*+]	t391	*inuA, tetK*	*tst1, egc, PVL*	*sak, chp, scn*	*agrIII, cap8*
	MRSA-IV	t019	*fusC, ermC*	*PVL, sea , egc*	*scn*	*agrIII*
	MRSA-IV + *SCCfus*	t267	*fusC*	*egc, sea*	*sak, chp*	*agrIII*
CC80	MRSA-IV [*PVL*+]	t502	*aphA3, msrA, mphC*	*PVL, etD*	*sak, chp*	*agrIII, cap8*
	MRSA-IV	t639	*aphA3, fusB*	*PVL, etD*	*sak, scn*	*agrIII, edinB*
CC8	MRSA-IV	t008	*blaZ, mphC*	*sek, seq*	*sak, scn*	*agrI, cap5*
	MRSA-IV	t450	*blaZ, ermC, fusC*	*egc, seb*	*sak, scn*	*agrI, cap5*
CC121	MRSA-V [*PVL*+]	t1192	*fusC, tetK, msrA*	*seb, PVL, egc*	*sak, scn*	*agrIV*
	MRSA-V	t757	*blaZ, tetK, fusC*	*seb, PVL*	*scn*	*agrIV*
	MRSA-V	t1991	*fusC, aacA-aphD*	*egc, sek*	*scn*	*agrIV*
CC152	MRSA-V [*PVL*+]	t008	*blaZ, fusC, aacA-aphD*	*sea, egc, PVL*	*sak, scn*	*agrI, cap5*
	MSSA [*PVL*+]	t002	*blaZ, fusC*	*egc, PVL*	*sak, scn*	*agrI*
CC361	MRSA-IV + *SCCfus*	t315	*msrA, mphC, ermC*	*tst1, egc*	*scn*	*agrI, cap8*
	MRSA-IV	t502	*blaZ, tetK, fusC*	*tst1*	*sak, scn*	*agrI*

CC clonal complexes, *MRSA* methicillin-resistant *Staphylococcus aureus*, *MSSA* methicillin-susceptible *Staphylococcus aureus*, *Spa typing* *Staphylococcus* protein A typing

via *dfrS1*, reflecting its adaptability to resist commonly prescribed antibacterial drugs. Complex CC15 is of particular interest due to its resistance to multiple classes, including aminoglycosides, macrolides, and tetracycline, primarily through genes such as *aacA-aphD* and *tetK*. The relatively high occurrence of tetracycline resistance in this complex could indicate a historical selection pressure, potentially from the widespread use of this antibacterial drug class in community settings. Similarly, CC97 isolates displayed significant resistance to fusidic acid and aminoglycosides, with the presence of *aacA-aphD* and *fusC*, which are both resistance determinants for gentamicin and fusidic acid, respectively (Table 1).

Toxin and virulence gene distribution

A notable feature across the MRSA isolates was the diversity of virulence genes, particularly those encoding toxins and immune-modulatory proteins. The *PVL* gene was highly prevalent, especially within CC5, CC30, and CC80 complexes, highlighting their enhanced potential for causing invasive infections, particularly skin and soft tissue infections. *PVL*'s association with increased virulence may contribute to the success of these complexes in both hospital and community settings, as evidenced by the rising frequency of *PVL*-positive CA-MRSA strains globally. The presence of *tst1*, the gene encoding toxic shock syndrome toxin-1, was mainly restricted to CC22 and CC30, with CC22 often associated with both *tst1* and *egc*, a gene cluster coding for multiple enterotoxins. This toxin combination is particularly concerning as it suggests an increased capacity for toxin-mediated disease in these complexes. Additionally, the high frequency of enterotoxins such as *sea* and *seb* in CC5 and CC15 further emphasizes the pathogenic potential of these MRSA strains. The presence of multiple enterotoxins could increase the risk of foodborne illnesses and other toxin-mediated syndromes if MRSA contaminates food handling areas.

Phage and accessory gene content

The variation in *agr* types may influence biofilm formation and the overall pathogenicity of these strains. Moreover, capsular polysaccharide types (*cap5* and *cap8*) observed in these isolates are known to contribute to immune evasion. The predominant detection of *cap5* in CC5 and CC22 isolates may enhance their resistance to phagocytosis, aiding their survival in hospital environments where immune pressures are high.

Immune-modulatory factors were prevalent, with most isolates carrying *sak* (staphylokinase), *chp* (chemotaxis inhibitory protein), and *scn* (staphylococcal complement inhibitor). These HLB-converting phages were widely distributed across various clonal complexes, particularly CC5, CC22, and CC97, enhancing the isolates' capacity to evade host immune defences. The high prevalence of these immune-modulatory genes highlights significant selective pressures in both hospital and community settings, favouring strains capable of persistence and recurrent infections. Accessory gene regulators (*agr*), especially *agr* types I and II, were frequently detected among the isolates. *Agr* type I was notably dominant in CC6 and CC22 isolates, whereas *agr* type II was primarily identified in CC5 and CC15, clonal complexes often associated with biofilm formation and chronic infection persistence. The distribution of *agr* types, combined with the presence of capsular polysaccharide genes (*cap5* and *cap8*), underscores the diverse regulatory strategies employed by these isolates. These mechanisms enable virulence factor expression, biofilm development, and immune evasion, contributing to their adaptability and survival in diverse host environments.

Antibacterial drug resistance patterns and genetic profiles in MRSA isolates

Table 2 categorizes the MRSA isolates based on their resistance to various antibacterial drug classes, highlighting both the extent of multi-drug resistance and the specific resistance combinations observed in the patient cohort. The isolates are grouped by the number of antibacterial drug classes they resist, from six classes down to single-drug resistance, allowing us to understand the spectrum of resistance within the MRSA population in the hospital setting.

Table 2 Antibacterial drug resistance profiles of MRSA isolates

S. No	No. of antibacterial drug classes	Resistance profile	No. of isolates
1.	6	PG, Km, Em, Clin, Cm, Tet	10
2.	6	PG, Em, Tet, Tp, Fa, Cip	8
3.	6	PG, Gn, Em, Tet, Fa, Cip	6
4.	5	PG, Gn, Km, Clin, Fa	12
5.	5	PG, Km, Clin, Tp, Fa	9
6.	5	PG, Em, Clin, Tp, Cip	14
7.	5	PG, Em, Clin, Tet, Cip	11
8.	4	PG, Km, Tet, Tp	13
9.	4	PG, Clin, Tet, Cip	15
10.	4	PG, Em, Tp, Cip	7
11.	3	PG, Gn, Fa	16
12.	3	PG, Tp, Fa	10
13.	2	PG, Em	18
14.	2	PG, Tp	12
15	1	PG	29

Cm chloramphenicol, *Cip* ciprofloxacin, *Clin* clindamycin, *Em* erythromycin, *Fa* fusidic acid, *Gn* gentamycin, *Km* kanamycin, *PG* penicillin G, *Tet* tetracycline

Highly multi-drug-resistant isolates (6 classes)

Isolates resistant to six antibacterial drug classes represent significant multi-drug resistance, often a hallmark of healthcare-associated MRSA strains that face high antibacterial drug exposure. These profiles include combinations like penicillin (PG), kanamycin (Km), erythromycin (Em), clindamycin (Clin), chloramphenicol (Cm), and tetracycline (Tet) and are seen in ten isolates. Additionally, resistance to trimethoprim (Tp), fusidic acid (Fa), and ciprofloxacin (Cip) appear in other six-class profiles, seen in eight and six isolates respectively, indicating variability in the resistance mechanisms across different MRSA lineages.

Five-class resistance patterns

Approximately 20% of isolates (five-class resistance) exhibit notable multi-drug resistance, commonly to penicillin, gentamicin (Gn), and kanamycin (Km), with additional resistance to clindamycin or fusidic acid. Profiles like PG, Km, Clin, Tp, Fa and PG, Em, Clin, Tp, and Cip are among the more frequent five-class combinations, comprising 9 to 14 isolates each. These isolates are of particular concern because they show substantial resistance to diverse antibacterial drug groups, limiting therapeutic options and potentially leading to prolonged infections.

Four-class and three-class resistance

Four-class resistant isolates (13 to 15 isolates per profile) are dominated by combinations of PG with either Km, Tet, or Tp. These profiles are prevalent and represent a transitional group between high- and moderate-resistance MRSA strains. Interestingly, isolates with three-class resistance, such as those resistant to PG, Tp, and Fa (ten isolates), maintain resistance but with a narrower scope, potentially showing a different resistance development pathway or lower selective pressure from multiple antibacterial drugs.

Lower-level resistance (two or fewer classes)

A sizable portion of the MRSA population shows limited resistance, with two-class and single-class resistance observed in the largest subsets (12 to 29 isolates). Two-class resistance profiles are often to PG with either Em or Tp, reflecting basic antibacterial drug resistance mechanisms common in MRSA strains. Notably, isolates resistant solely to PG comprise the largest single category (29 isolates), showing typical MRSA b*etA*-lactam resistance without additional multi-drug resistance traits.

Occurrence of virulence determinants in MRSA isolates

The analysis of 190 MRSA isolates revealed a high prevalence of virulence-associated genes, notably those encoding for toxins, immune-modulatory proteins, and other virulence factors that enhance pathogen adaptability in hospital environments. The *PVL* gene, an important marker for cytotoxicity, was detected in 20% of the isolates, predominantly within CC5, CC80, and CC30 clonal complexes. Among enterotoxins, 50% of isolates carried the enterotoxin gene cluster (*egc*), which includes genes for several enterotoxins (*seg*, *sei*, *selm*, *seln*, *selo*, *selu*). Additional enterotoxin genes, such as *sea, seb, sed,* and *ser,* were present at varying frequencies, with *sea* and *seb* being most prevalent, detected in 35% and 25% of isolates, respectively. The toxic shock syndrome toxin-1 (*tst1*) gene was found in 15% of the isolates, primarily within CC22 and CC30, clonal complexes often implicated in healthcare-associated MRSA infections. The presence of *tst1* in these complexes, combined with multi-drug resistance patterns, could elevate the risk of severe, toxin-mediated syndromes in the hospital population. Exfoliative toxin genes, including *etA* and *etD*, were detected at lower frequencies, aligning with their sporadic occurrence in MRSA but suggesting a potential role in specific infection presentations when present.

Immune-modulatory factors were common, with the majority of isolates harbouring *sak* (staphylokinase), *chp* (chemotaxis inhibitory protein), and *scn* (staphylococcal complement inhibitor). These HLB-converting phages were widespread across different clonal complexes, particularly in CC5, CC22, and CC97, enhancing the isolates' ability to evade host immune responses. This high prevalence of immune-modulatory genes suggests strong selective pressures in both hospital and community environments, promoting strains that can persist and cause recurrent infections. Accessory gene regulators, particularly *agr* types I and II, were frequently observed across the isolates. *Agr* type I was prominent among CC6 and CC22 isolates, whereas *agr* type II was predominantly found in CC5 and CC15, which are typically associated with biofilm formation and infection persistence. The distribution of *agr* types, along with capsular polysaccharide genes (*cap5* and *cap8*), further indicates that these isolates possess varied regulatory mechanisms for virulence factor expression, biofilm formation, and immune evasion, contributing to their adaptability and survival across different host environments.

Discussion

This study elucidates the molecular and genetic landscape of methicillin-resistant *Staphylococcus aureus* (MRSA) isolates at KKH Hospital in Saudi Arabia, focusing on antimicrobial resistance, genetic diversity, and virulence potential. The findings highlight the complex epidemiology of MRSA in healthcare settings and reveal patterns that align with both local healthcare-associated (HA) MRSA dynamics and broader global trends. These insights provide crucial evidence for enhancing infection control and antimicrobial stewardship programs in the region. The high prevalence of beta-lactam resistance, confirmed via cefoxitin susceptibility testing, aligns with findings globally, where resistance to penicillin and related drugs remains a hallmark of MRSA. Notably, 70% of isolates exhibited resistance to fusidic acid, and 55% to ciprofloxacin, likely reflecting local prescribing practices and selective pressures within the hospital. MDR was identified in 78% of isolates, consistent with global trends, and highlights MRSA's adaptability to diverse antimicrobial agents. This is particularly concerning given the reliance on last-resort drugs like vancomycin and teicoplanin for MDR MRSA infections.

The study identified diverse clonal complexes (CCs), with CC5, CC6, and CC22 being the most prevalent. The dominance of CC5 and CC6 aligns with their global recognition as successful community-associated MRSA (CA-MRSA) lineages that have transitioned into healthcare settings. The diversity within these CCs, including *spa* types like t304 (CC5) and t688 (CC6), indicates multiple routes of MRSA introduction and adaptation. The detection of less common lineages, such as CC15, CC30, and CC80, suggests sporadic introductions likely from community or international sources. This genetic variability underscores the need for continuous surveillance to monitor changes in MRSA population dynamics and resistance trends.

The virulence potential of these MRSA isolates is underscored by the high prevalence of toxin-encoding genes. The *PVL* gene, particularly in CC5, CC30, and CC80, is associated with severe skin and soft tissue infections and poses a significant challenge in hospital settings due to its enhanced transmissibility and infection severity (Szumlanski et al.

2022). Additionally, enterotoxin genes (*sea* and *seb*) were frequently detected, raising concerns about toxin-mediated diseases in immunocompromised patients. The presence of the *tst1* gene (toxic shock syndrome toxin-1), primarily in CC22 and CC30, highlights the risk of severe toxin-mediated conditions such as toxic shock syndrome (Atchade et al. 2024).

Immune-evasion determinants like *sak*, *chp*, and *scn*, largely associated with HLB-converting phages, were widespread across CCs, facilitating MRSA's ability to evade host immune responses. This immune evasion likely contributes to persistence and recurrent infections. Variations in accessory gene regulators (*agr* types I and II) and capsular polysaccharide genes (*cap5* and *cap8*) further demonstrate MRSA's adaptability (Howden et al. 2023). *Agr* type I predominated in CC6 and CC22, while *agr* type II was common in CC5 and CC15, lineages associated with biofilm formation and chronic infections. These traits emphasize the importance of targeting immune evasion mechanisms in infection control strategies (Lai et al. 2024).

The study's findings have significant clinical implications. The high prevalence of MDR strains underscores the need for judicious antimicrobial use and tailored stewardship programs. The detection of highly transmissible and virulent clonal complexes, such as CC5 and CC22, highlights the importance of molecular typing and genomic surveillance in monitoring MRSA transmission and emerging strains. Future studies incorporating longitudinal and whole-genome analyses are essential to track the evolution of resistance and virulence profiles, offering valuable insights to guide infection control and therapeutic approaches (Almutairy 2024).

Conclusion

In summary, this study offers a detailed analysis of the molecular and genetic features of MRSA isolates from a Saudi Arabian hospital. The results highlight the intricate nature of MRSA epidemiology in healthcare settings, showcasing the interplay between antimicrobial resistance, genetic diversity, and virulence. The identification of dominant clonal complexes, along with their resistance and virulence profiles, underscores the critical need for ongoing surveillance and molecular characterization to guide targeted infection control measures and antimicrobial stewardship programs.

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Data availability All source data for this work (or generated in this study) are available upon reasonable request.

Declarations

Clinical trial number Not applicable.

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